

Transform Table and Trigonometric Identities

Allowed in exams of course Signals and System without any handwritten notes.

Z-Transform Pairs

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$x[n]$	$X[z]$
$\delta[n - k]$	z^{-k}
$u[n]$	$\frac{z}{z - 1}$
$nu[n]$	$\frac{z}{(z - 1)^2}$
$n^2u[n]$	$\frac{z(z + 1)}{(z - 1)^3}$
$n^3u[n]$	$\frac{z(z^2 + 4z + 1)}{(z - 1)^4}$
$\gamma^n u[n]$	$\frac{z}{z - \gamma}$
$\gamma^{n-1} u[n - 1]$	$\frac{1}{z - \gamma}$
$n\gamma^n u[n]$	$\frac{\gamma z}{(z - \gamma)^2}$
$n^2\gamma^n u[n]$	$\frac{\gamma z(z + \gamma)}{(z - \gamma)^3}$
$\frac{n(n - 1)(n - 2) \cdots (n - m + 1)}{\gamma^m m!} \gamma^n u[n]$	$\frac{z}{(z - \gamma)^{m+1}}$
$ \gamma ^n \cos \beta n u[n]$	$\frac{z(z - \gamma \cos \beta)}{z^2 - (2 \gamma \cos \beta)z + \gamma ^2}$
$ \gamma ^n \sin \beta n u[n]$	$\frac{z \gamma \sin \beta}{z^2 - (2 \gamma \cos \beta)z + \gamma ^2}$
$r \gamma ^n \cos(\beta n + \theta) u[n]$	$\frac{rz[z \cos \theta - \gamma \cos(\beta - \theta)]}{z^2 - (2 \gamma \cos \beta)z + \gamma ^2}$

Z-Transform Properties

Operation	$x[n]$	$X[z]$
Addition	$x_1[n] + x_2[n]$	$X_1[z] + X_2[z]$
Scalar multiplication	$ax[n]$	$aX[z]$
Right shifting	$x[n-m]u[n-m]$	$\frac{1}{z^m}X[z]$
	$x[n-m]u[n]$	$\frac{1}{z^m}X[z] + \frac{1}{z^m} \sum_{n=1}^m x[-n]z^n$
	$x[n-1]u[n]$	$\frac{1}{z}X[z] + x[-1]$
	$x[n-2]u[n]$	$\frac{1}{z^2}X[z] + \frac{1}{z}x[-1] + x[-2]$
	$x[n-3]u[n]$	$\frac{1}{z^3}X[z] + \frac{1}{z^2}x[-1] + \frac{1}{z}x[-2] + x[-3]$
Left shifting	$x[n+m]u[n]$	$z^mX[z] - z^m \sum_{n=0}^{m-1} x[n]z^{-n}$
	$x[n+1]u[n]$	$zX[z] - zx[0]$
	$x[n+2]u[n]$	$z^2X[z] - z^2x[0] - zx[1]$
	$x[n+3]u[n]$	$z^3X[z] - z^3x[0] - z^2x[1] - zx[2]$
Multiplication by γ^n	$\gamma^n x[n]u[n]$	$X\left[\frac{z}{\gamma}\right]$
Multiplication by n	$nx[n]u[n]$	$-z \frac{d}{dz} X[z]$
Time reversal	$x[-n]$	$X[1/z]$
Time convolution	$x_1[n] * x_2[n]$	$X_1[z]X_2[z]$
Initial value	$x[0]$	$\lim_{z \rightarrow \infty} X[z]$
Final value	$\lim_{N \rightarrow \infty} x[N]$	$\lim_{z \rightarrow 1} (z-1)X[z]$ Poles of $(z-1)X[z]$ inside the unit circle

Laplace Transform Pairs

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt$$

$x(t)$	$X(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
$tu(t)$	$\frac{1}{s^2}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$
$e^{\lambda t} u(t)$	$\frac{1}{s - \lambda}$
$te^{\lambda t} u(t)$	$\frac{1}{(s - \lambda)^2}$
$t^n e^{\lambda t} u(t)$	$\frac{n!}{(s - \lambda)^{n+1}}$
$\cos bt u(t)$	$\frac{s}{s^2 + b^2}$
$\sin bt u(t)$	$\frac{b}{s^2 + b^2}$
$e^{-at} \cos bt u(t)$	$\frac{s + a}{(s + a)^2 + b^2}$
$e^{-at} \sin bt u(t)$	$\frac{b}{(s + a)^2 + b^2}$
$re^{-at} \cos(bt + \theta) u(t)$	$\frac{(r \cos \theta)s + (a r \cos \theta - b r \sin \theta)}{s^2 + 2as + (a^2 + b^2)}$

Laplace Transform Properties

Operation	$x(t)$	$X(s)$
Addition	$x_1(t) + x_2(t)$	$X_1(s) + X_2(s)$
Scalar multiplication	$kx(t)$	$kX(s)$
Time differentiation	$\frac{dx(t)}{dt}$	$sX(s) - x(0^-)$
	$\frac{d^2x(t)}{dt^2}$	$s^2X(s) - sx(0^-) - \dot{x}(0^-)$
	$\frac{d^3x(t)}{dt^3}$	$s^3X(s) - s^2x(0^-) - s\dot{x}(0^-) - \ddot{x}(0^-)$
	$\frac{d^nx(t)}{dt^n}$	$s^nX(s) - \sum_{k=1}^n s^{n-k}x^{(k-1)}(0^-)$
Time integration	$\int_{0^-}^t x(\tau) d\tau$	$\frac{1}{s}X(s)$
	$\int_{-\infty}^t x(\tau) d\tau$	$\frac{1}{s}X(s) + \frac{1}{s} \int_{-\infty}^{0^-} x(t) dt$
Time shifting	$x(t - t_0)u(t - t_0)$	$X(s)e^{-st_0} \quad t_0 \geq 0$
Frequency shifting	$x(t)e^{s_0t}$	$X(s - s_0)$
Frequency differentiation	$-tx(t)$	$\frac{dX(s)}{ds}$
Frequency integration	$\frac{x(t)}{t}$	$\int_s^\infty X(z) dz$
Scaling	$x(at), a \geq 0$	$\frac{1}{a}X\left(\frac{s}{a}\right)$
Time convolution	$x_1(t) * x_2(t)$	$X_1(s)X_2(s)$
Frequency convolution	$x_1(t)x_2(t)$	$\frac{1}{2\pi j}X_1(s) * X_2(s)$
Initial value	$x(0^+)$	$\lim_{s \rightarrow \infty} sX(s) \quad (n > m)$
Final value	$x(\infty)$	$\lim_{s \rightarrow 0} sX(s) \quad [\text{poles of } sX(s) \text{ in LHP}]$

Fourier Transform Pairs

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Note: $a > 0$, wherever it appears in the table

$x(t)$	$X(\omega)$
$e^{-at}u(t)$	$\frac{1}{a + j\omega}$
$e^{at}u(-t)$	$\frac{1}{a - j\omega}$
$e^{-a t }$	$\frac{2a}{a^2 + \omega^2}$
$te^{-at}u(t)$	$\frac{1}{(a + j\omega)^2}$
$t^n e^{-at}u(t)$	$\frac{n!}{(a + j\omega)^{n+1}}$
$\delta(t)$	1
1	$2\pi\delta(\omega)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin \omega_0 t$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$
$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
$\text{sgn } t$	$\frac{2}{j\omega}$
$\cos \omega_0 t u(t)$	$\frac{\pi}{2}[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] + \frac{j\omega}{\omega_0^2 - \omega^2}$
$\sin \omega_0 t u(t)$	$\frac{\pi}{2j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] + \frac{\omega_0}{\omega_0^2 - \omega^2}$
$e^{-at} \sin \omega_0 t u(t)$	$\frac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$
$e^{-at} \cos \omega_0 t u(t)$	$\frac{a + j\omega}{(a + j\omega)^2 + \omega_0^2}$
$\text{rect}\left(\frac{t}{\tau}\right)$	$\tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$
$\frac{W}{\pi} \text{sinc}(Wt)$	$\text{rect}\left(\frac{\omega}{2W}\right)$

Fourier Transform Properties

Operation	$x(t)$	$X(\omega)$
Scalar multiplication	$kx(t)$	$kX(\omega)$
Addition	$x_1(t) + x_2(t)$	$X_1(\omega) + X_2(\omega)$
Conjugation	$x^*(t)$	$X^*(-\omega)$
Duality	$X(t)$	$2\pi x(-\omega)$
Scaling (a real)	$x(at)$	$\frac{1}{ a } X\left(\frac{\omega}{a}\right)$
Time shifting	$x(t - t_0)$	$X(\omega)e^{-j\omega t_0}$
Frequency shifting (ω_0 real)	$x(t)e^{j\omega_0 t}$	$X(\omega - \omega_0)$
Time convolution	$x_1(t) * x_2(t)$	$X_1(\omega)X_2(\omega)$
Frequency convolution	$x_1(t)x_2(t)$	$\frac{1}{2\pi} X_1(\omega) * X_2(\omega)$
Time differentiation	$\frac{d^n x(t)}{dt^n}$	$(j\omega)^n X(\omega)$
Time integration	$\int_{-\infty}^t x(u) du$	$\frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$
Parseval's Theorem	$E_x = \int_{-\infty}^{\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) ^2 d\omega$	

TRIGONOMETRY

LAWS AND IDENTITIES

TANGENT IDENTITIES	RECIPROCAL IDENTITIES	PYTHAGOREAN IDENTITIES	PERIODIC IDENTITIES
$\tan \theta = \frac{\sin \theta}{\cos \theta}$ $\cot \theta = \frac{\cos \theta}{\sin \theta}$	$\csc \theta = \frac{1}{\sin \theta}$ $\sec \theta = \frac{1}{\cos \theta}$ $\cot \theta = \frac{1}{\tan \theta}$	$\sin^2 \theta + \cos^2 \theta = 1$ $\tan^2 \theta + 1 = \sec^2 \theta$ $\cot^2 \theta + 1 = \csc^2 \theta$	$\sin(\theta + 2\pi n) = \sin \theta$ $\cos(\theta + 2\pi n) = \cos \theta$ $\tan(\theta + \pi n) = \tan \theta$ $\csc(\theta + 2\pi n) = \csc \theta$ $\sec(\theta + 2\pi n) = \sec \theta$ $\cot(\theta + \pi n) = \cot \theta$
EVEN/ODD IDENTITIES	DOUBLE ANGLE IDENTITIES	HALF ANGLE IDENTITIES	LAW OF COSINES
$\sin(-\theta) = -\sin \theta$ $\cos(-\theta) = \cos \theta$ $\tan(-\theta) = -\tan \theta$ $\csc(-\theta) = -\csc \theta$ $\sec(-\theta) = \sec \theta$ $\cot(-\theta) = -\cot \theta$	$\sin(2\theta) = 2 \sin \theta \cos \theta$ $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$ $= 2 \cos^2 \theta - 1$ $= 1 - 2 \sin^2 \theta$ $\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$	$\sin\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 - \cos \theta}{2}}$ $\cos\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 + \cos \theta}{2}}$ $\tan\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$	$a^2 = b^2 + c^2 - 2bc \cos \alpha$ $b^2 = a^2 + c^2 - 2ac \cos \beta$ $c^2 = a^2 + b^2 - 2ab \cos \gamma$

PRODUCT TO SUM IDENTITIES	SUM TO PRODUCT IDENTITIES
$\sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$ $\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$ $\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$ $\cos \alpha \sin \beta = \frac{1}{2} [\sin(\alpha + \beta) - \sin(\alpha - \beta)]$	$\sin \alpha + \sin \beta = 2 \sin\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right)$ $\sin \alpha - \sin \beta = 2 \cos\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right)$ $\cos \alpha + \cos \beta = 2 \cos\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right)$ $\cos \alpha - \cos \beta = -2 \sin\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right)$